IILM UNIVERSITY, GREATER NOIDA

Data Analyst Internship at BeeSkilled

A Professional 6-Week B.Tech CSE Internship Presentation Report

INTERNSHIP DURATION:
30 July 2026 – 10 September 2026

CANDIDATE CREDENTIALS

Anshika Singh

Roll Number: 25SCS1003004925
Academic Batch: 2026–27
Degree & Program: B.Tech (Computer Science & Engineering)
Department: School of Computer Science & Engineering
University: IILM University, Greater Noida, U.P.

HOST ORGANIZATION & EVALUATION

BeeSkilled (Online Learning & Skill Development)

Industry Alignment: Registered with AICTE & MSME

Faculty/Mentor: Ms. Simranjeet Kaur (Founder & Program Head)

BeeSkilled: Aligning Academic Foundations with Practical Skill Development

Host Organization Profile

- **Company Name & Brand:** BeeSkilled — operating with the core tagline "Tech and Internships".
- **Focus Area:** An online platform focused on structured, project-oriented internship and skill-development programs for engineering and technology students.
- **Industry Credentials:** Fully registered and recognized by AICTE (All India Council for Technical Education) and MSME (Ministry of Micro, Small, and Medium Enterprises).
- **Strategic Objective:** Strengthen practical skills by providing structured coursework, practical assignments, and end-to-end industry capstones to bridge the academia-industry gap.

Internship Program Overview

- **Specialization Domain:** Data Analyst
- **Program Duration:** 6 Weeks (30 July 2026 – 10 September 2026)
- **Mode of Internship:** Online Structured Internship with Weekly Hands-on Deliverables
- **Evaluation & Guidance:** Ms. Simranjeet Kaur, Founder & Program Head, BeeSkilled
- **Program Structure:** Weekly learning modules coupled with practical assignments and an intensive, data-driven End-to-End Business Analytics Capstone Project.
- **Key Methodology:** Transforming raw datasets through Microsoft Excel, Python/Pandas scripts, and Power BI visualization models to generate stakeholder business advice.

Solving the Business Dilemma: Extracting Actionable Insights from Raw Data

The Business & Data Challenge

In modern commerce, organizations produce massive volumes of transactional data through their daily activities. However, this raw data is usually chaotic, containing duplicate records, missing entries, inconsistent date formatting, and unstructured columns. Raw data alone is incapable of supporting business decisions. The analytical challenge addressed during this internship is to define how unprocessed transactional datasets can be systematically parsed, validated, cleaned, modeled, and visualized to yield trustworthy, executive-ready insights that prevent profit leakage.

Core Internship Objectives

- Fundamental Competency: Master the standard 6-stage data analytics workflow.
- Spreadsheet Mechanics: Ingest and aggregate large-scale corporate data in Microsoft Excel using Pivot Tables and SUM-based aggregation.
- Programmatic Wrangling: Build robust Python scripts using the Pandas library to load datasets, evaluate shape/info, detect duplicates, and treat missing values.
- Business Intelligence: Develop rich, interactive Power BI dashboards and construct calculated DAX measures for critical business KPI tracking.

Scope of Practical Project Work

- Structured Datasets: Focuses on working with multi-column, structured sales, order-related, and performance-related datasets.
- Microsoft Excel Analytics: Processing a 51,290-record Global Superstore Orders dataset to extract product-category and segment trends.
- Python & Pandas Wrangling: Inspecting sales records and cleaning inconsistent workout datasets with missing entries and duplicate rows.
- End-to-End Power BI Capstone: Validating a 700-record sales/financial dataset, defining custom KPIs with DAX, and formulating executive-level strategic recommendations.

Leveraging an Integrated Analytical Toolchain across the 6-Stage Workflow

Tool / Technology	Primary Purpose	Practical Application in Internship Program
Microsoft Excel	Spreadsheet-based data organization and analysis	Built pivot-table dashboards to aggregate total revenue, category-wise performance, yearly sales trends, and segment revenue on a 51,290-record Superstore dataset.
Python Programming	General-purpose language for data-analysis scripting	Constructed scripts to handle dataset imports, general structural examinations, and missing-value treatments across Weeks 2 and 3.
Pandas Library	Structured data manipulation and tabular analysis	Utilized shape, info(), head(), and describe() for 200-row dataset inspection. Standardized formats, interpolated date fields, and imputed calories in a 32-row dataset.
Power BI	Enterprise business intelligence and visualization	Developed an interactive business dashboard. Created custom calculated DAX measures for core KPIs (Total Sales, Total Profit, Profit Margins, Loss-Making Orders).

THE RECURRING 6-STAGE DATA ANALYTICS WORKFLOW:

Step 1
Raw Data
Load files

Step 2
Cleaning
Fix issues

Step 3
Transform
Derive fields

Step 4
Analysis
Run scripts

Step 5
Visuals
Build dashboards

Step 6
Insights
Recommend

Pivot Table Sales Performance on the 51,290-Row Global Superstore Dataset

Dataset Ingestion & Sales Trends

The Global Superstore Orders dataset is a large transactional spreadsheet consisting of approximately 51,290 records and 24 columns. Standard Excel Pivot Tables were used to construct aggregation and sales-performance trend summaries across years and quarters.

Yearly Sales Performance Trend (Steady Growth):

- **Calendar Year 2012:** $2.26 Million
- **Calendar Year 2013:** $2.68 Million (▲ +18.5%)
- **Calendar Year 2014:** $3.41 Million (▲ +27.2%)
- **Calendar Year 2015:** $4.30 Million (▲ +26.1%)

Category & Customer Segment Analysis

Product Category Sales (Technology Leading):

- **Technology Sales:** $4.74 Million (Highest performer)
- **Furniture Sales:** $4.11 Million
- **Office Supplies Sales:** $3.79 Million

Customer Segment Performance (Consumer Dominating):

- **Consumer Segment:** $6.51 Million (Highest revenue share)
- **Corporate Segment:** $3.82 Million
- **Home Office Segment:** $2.31 Million

Initial Dataset Auditing and Summary Statistics using Python and Pandas

Dataset Structure & Pandas Inspection

A sales dataset containing 200 transactional records and 10 columns was loaded into a Python DataFrame to inspect column types, dimensions, and contents before executing transformations.

- **df.shape Method:** Evaluates the dimensions of the file. Returned exactly 200 records and 10 column attributes.
- **df.info() Method:** Verifies column naming, indices, and checks non-null counts against total row indices to spot missing data patterns.
- **df.head() Method:** Previews the first 5 records of raw transactional data, verifying columns like order_id, product_name, and total_price.
- **df.columns Attribute:** Returns an array of clean, verified header strings, validating fields (order_date, category, quantity, unit_price, etc.)

Central Tendencies & Descriptive Stats

The df.describe() method was executed on numeric columns to immediately summarize the central tendency, spread, standard deviation, percentiles, and maximum boundaries of variables.

Key Statistical Metrics from df.describe() Output:

- **Mean Order Quantity:** 4.91 Units per sales order transaction
- **Mean Unit Price:** $2,421.50 per individual item
- **Mean Total Price:** $12,101.54 per order transaction
- **Maximum Total Price:** $47,940.00 (Single highest transaction in dataset)
- **Standard Deviation:** Provides statistical spread of product pricing across regions.

Wrangling, Interpolation, and Field Derivation on a 32-Row Workout Dataset

Inconsistent Raw Data State

The workout/activity dataset consisted of 32 rows and 5 columns: Duration, Date, Pulse, Maxpulse, and Calories. The initial programmatic audit identified several critical data-quality anomalies:

- **Duplicate Entries:** One identical duplicate record was flagged in the dataset, risking statistical bias in calculations.
- **Missing Date Field:** The 'Date' column contained one missing entry, and some date values were stored in non-standard, inconsistent text formats.
- **Missing Numeric Metrics:** The 'Calories' column contained two blank entries, which would break sum or mean calculations.

Pandas Data Cleaning & Wrangling

- **Date Standardisation:** Converted the Date field into a uniform pandas datetime format.
- **Interpolation treatment:** The single missing Date field was programmatically resolved using forward-fill/interpolation techniques.
- **Mean Imputation:** The two missing Calories entries were replaced using the mean value calculated from the rest of the column, preserving statistical weight.
- **Duplicate Removal:** Removed the single duplicate row, resulting in exactly 31 clean unique rows.
- **Derived Column:** Generated an analytical column: Calories_Per_Minute = Calories / Duration for detailed rate evaluation.
- **File Outputs:** Saved the wrangled, 6-column dataset as cleaned_data.csv. Filtered and exported high-intensity workout records (>300 Calories) as filtered_data.csv.

Consolidating Wrangling, Modeling, and Storytelling on Financial Sales Data

Week 4 merged all technical competencies into an end-to-end commercial business analytics project. The objective was to apply the full 6-stage analytics workflow to a transactional sales dataset containing 700 records and 16 original attributes.

01

Phase 1: Validation & Cleaning

Inspecting raw fields, standardizing empty boundaries, validating discounts, and auditing duplicate rows.

02

Phase 2: Power BI / DAX Modeling

Constructing a robust tabular model, establishing relational continuity, and generating custom calculated business measures.

03

Phase 3: Visual Storytelling

Building interactive executive dashboards to track segment profit leakage, country sales distribution, and product trends.

Overcoming Data Gaps and Engineering Business Intelligence Measures

Capstone Data Validation

Before loading the capstone transaction file (700 rows, 16 columns) into Power BI, a thorough data quality audit was conducted to preserve statistical accuracy.

- **Blank Value Detection:** Found exactly 53 records with empty cell entries in the 'Discount Band' categorical column.
- **Logical Evaluation & Treatment:** Cross-reference check showed these 53 transactions had a corresponding discount value of exactly zero. Instead of removing the rows (which would delete valid transactions), they were categorized as 'No Discount'.
- **Duplicate Auditing:** A full unique-constraint check was run, and zero duplicate rows were identified, confirming the dataset was ready for BI ingestion.

DAX Key Performance Measures

Custom business calculations were formulated using DAX (Data Analysis Expressions) to establish high-level KPIs and filter conditions for interactive reports.

- **Total Sales Measure:** SUM of sales transaction amounts to represent top-line gross intake.
- **Total Profit Measure:** SUM of net profit margins computed across transaction rows.
- **Total Units Sold:** Aggregated sum of units distributed globally across segments.
- **Profit Margin Measure:** Divides total profits by total sales to evaluate overall operational cost-efficiency.
- **Average Order Value:** Identifies the average sales size per order row to trace size behaviors.
- **Loss-Making Orders Count:** Counts records where profit value fell below zero to identify financial leakage.

Top-Line Metrics and Profit Leakage Traced to Segment and Discount Levels

TOTAL CAPSTONE SALES
$118,726,350
Top-line commercial revenue

TOTAL CAPSTONE PROFIT
$16,893,702
Net commercial profit

UNITS DISTRIBUTED
1,125,806
Total product items sold

COMMERCIAL MARGIN
14.2%
Healthy overall profitability

Critical Analytical Discoveries & Performance Disparities

- **Segment Profitability Analysis:** Government was identified as the most profitable sector, generating approximately $11.39 Million in profits. In contrast, the Enterprise segment was identified as the ONLY loss-making sector, generating a net overall loss of -$614,546, indicating immediate operational failure.
- **Geographical Performance:** The United States of America (USA) achieved the highest geographical sales volume among all nations analyzed, contributing a total revenue of approximately $25.03 Million.
- **Product Performance Breakdown:** Paseo recorded the strongest overall metrics, leading both sales volume and profitability. Conversely, Carretera was identified as the weakest product, ranking lowest in both gross revenue and net profits.
- **Loss-Making Orders & Leakage:** Out of 700 total orders, exactly 58 transactions (8.3%) resulted in direct losses. Cross-tabulation showed these unprofitable sales heavily converged in the Medium and High discount bands, proving a correlation between deep discounts and profit erosion.
- **Cyclical Seasonality trends:** A study of monthly patterns showed that October and December 2014 achieved the highest historical sales volumes, whereas September 2013 registered the lowest sales performance during the studied timeline.

Guiding Stakeholder Decisions and Reviewing Core Analytical Capabilities

Strategic Recommendations

- **Reassess Discount Policies:** Immediately audit and tighten discount approval rules for Medium and High bands to stop profit leakage from loss-making orders (8.3% of total volume).
- **Establish Margin Thresholds:** Define minimum net profit-margin triggers in the visual dashboard to block unprofitable orders before transaction approval.
- **Investigate Enterprise Deficit:** Conduct a detailed operational audit into the Enterprise segment to diagnose why it produced a -$614,546 net loss despite high volumes.
- **Product Structure Review:** Audit pricing and COGS structures of the lowest-performing product (Carretera) to restore healthy segment margins.
- **Optimize Resource Planning:** Utilize identified cyclical seasonality to plan inventory, warehousing, and staff around high-demand windows in October and December.

Key Internship Learning Outcomes

- **Multi-Tool Proficiency:** Gained practical competency in Excel (Pivot Tables), Python scripting, Pandas dataframes, and Power BI DAX modeling.
- **Wrangling & Cleaning Expertise:** Acquired hands-on experience detecting duplicate rows, identifying blank records, standardizing date strings, and using mean/interpolation imputations.
- **Business Metric Design:** Developed ability to model financial KPIs (Gross Margin, Profit Margins, Average Order Values) rather than just parsing raw metrics.
- **Visual Storytelling & Business Advice:** Learned to build clean, impact-oriented visual interfaces and translate complex charts into actionable recommendations for business executives.

OFFICIAL VERIFICATION

Official Verification of 6-Week Data Analyst Internship Completion

Credential Details & Authenticity

This slide verifies the completion of the 6-week online internship program. The credentials represent completed course milestones, submitted assignments, and a successfully reviewed capstone dashboard.

- **Internship Domain:** Data Analyst
- **Issuing Organization:** BeeSkilled (Registered with AICTE & MSME)
- **Recipient Student Name:** Anshika Singh
- **Completed Duration:** 6 Weeks (30 July 2026 – 10 September 2026)
- **Official Issue Date:** 28 August 2026
- **Authorized Signature:** Simranjeet Kaur (Founder & Program Head)

Thank You!

An Professional Data Analyst Internship Completion Presentation

Presented By:

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